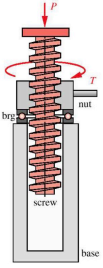


Ex3) Self-locking or back-driving



- For the screw jack in Ex1),
 - determine if is self-locking or not
 - compute the efficiency
 - Parameters from Ex1)
 - single-start thread,
 - the pitch $p = 0.2$ in,
 - the pitch diameter $d_p = 1.15$ in,
 - the collar diameter $d_c = 2$ in,
 - the lead screw friction $\mu = 0.15$,
 - the collar friction $\mu_B = 0.02$



- Verify that the scissor jack of ex2) is self-locking

Screw jack:

$$n_s = 1 \Rightarrow L = p = 0.2$$

$$\tan \lambda = \frac{L}{\pi d_p} = \frac{0.2}{\pi \cdot 1.15} = 0.055 < 0.15 = \mu$$

$$\text{Efficiency } \eta = \frac{1 - \mu \tan \lambda}{1 + \mu \cot \lambda} = \frac{1 - 0.15 \cdot 0.055}{1 + 0.15 \cdot \frac{1}{0.055}} = 0.2661 \quad (\text{no roller friction})$$

With roller friction:

$$\eta = \frac{PL}{2\pi T_{up}} = \frac{1000 \times 6.2}{2\pi \times 139.07} = 0.2289$$

Scissor jack:

$$\tan \lambda = \frac{L}{\pi d_p} = 0.0433 < \mu = 0.15$$

$\Rightarrow$ self-locking

$$T_{down} > 0 \Leftrightarrow \text{self-locking}$$